

TITLE: Collaborative Study Material and Online Quiz Platform

Members: 1. Sanjith K S (PES1UG24CS613)
2. Fathima Zahra (PES1UG24CS664)
3. Dhanush S Shekhar (PES1UG24CS662)

Requirement Elicitation

1. Peer-to-Peer Study Material Sharing

- Users shall be able to upload and share study materials that can be accessed by other users.
- Materials shall be organized based on subjects, domains, and topics for easy discovery.
- Users shall be able to view and download materials shared by peers.

2. Quiz Generation and Participation

- The system shall provide quizzes based on **shared study materials as well as general topics**.
- Users shall be able to attempt quizzes with different difficulty levels and question types.
- Quiz results and attempts shall be recorded for future performance analysis.

3. Collaborative Study Rooms

- Users shall be able to create or join study rooms based on a particular subject, domain, or topic.

- Users shall be able to upload and share external resources such as links, PDFs, videos, and reference materials within a study room.
- Resources shared in a study room shall be accessible to its members or all users, depending on the room's access settings.

4. Collaborative Question Banks

- The system shall provide question banks generated from **shared study materials as well as additional online sources**.
- Users shall be able to contribute, search, filter, and practice questions based on subject, topic, and difficulty.
- Questions from the question bank shall be usable for creating peer-based and LLM-generated quizzes.

5. LLM-Based Performance Analysis and Recommendations

- The LLM shall analyze quiz performance to identify the user's strengths, weaknesses, and areas requiring improvement.
- The LLM shall review relevant study materials and suggest suitable resources based on the user's performance.
- The system shall provide personalized recommendations for further study and practice.

6. Peer Review and Rating of Study Materials

- Users shall be able to rate and review study materials shared by other users.
- The system shall use peer feedback to indicate the quality or usefulness of materials.

- Users shall be able to report inaccurate, irrelevant, duplicate, or inappropriate materials.

7. Personalized User Material List

- Each user shall have a personal list where they can save or organize study materials they want to access later.
- Users shall be able to add LLM-recommended materials to their personal list.
- The personal list shall allow users to revisit selected materials and track their learning resources.

8. User Streaks and Reputation System

- The system shall track user activity and maintain learning streaks based on regular participation.
- Users shall earn reputation points for activities such as sharing useful materials, contributing questions, creating quizzes, and helping peers.
- The system shall provide **badges/achievements** based on milestones, contributions, and reputation.

9. Peer Validation and Content Recommendation

- Users shall be able to validate the usefulness and accuracy of peer-contributed materials and questions.
- The system shall use peer ratings, reviews, and usage patterns to identify high-quality content.
- The system shall recommend highly rated and relevant peer content based on the user's interests and performance.

Functional Requirements

FR-01: User Registration & Authentication

- The system shall allow users to register and securely log in.
- The system shall authenticate users before accessing protected features.
- The system shall support different user permissions based on their role.

FR-02: User Profile Management

- Users shall be able to view and update their profile information.
- The system shall maintain each user's contribution and activity history.
- Users shall be able to view their reputation and achievements.

FR-03: Channel Creation & Classification

- Users shall be able to create **Normal Channels** for domains, subjects, topics, or study groups.
- Authorized users/admins shall be able to create or approve **Authorized Channels**.
- The system shall clearly distinguish Authorized and Normal Channels.

FR-04: Channel Membership

- Users shall be able to search, join, and leave available channels.
- The system shall maintain channel membership information.
- Channel creators/moderators shall be able to manage members where applicable.

FR-05: Authorized Channel Management

- Authorized users shall be able to manage official study materials, question banks, and quizzes.
- Authorized channels shall support verified/official educational content.
- Authorized users shall have additional moderation and management privileges.

FR-06: Normal Channel Management

- Students shall be able to create and manage peer-based study channels.
- Members shall be able to contribute materials, questions, quizzes, and discussions.
- Content shall be community-driven and subject to peer rating/reporting.

FR-07: Peer-to-Peer Study Material Sharing

- Users shall be able to upload and share notes, PDFs, presentations, links, and other study resources.
- Materials shall be categorized by subject, topic, semester, and material type.
- Users shall be able to view and download materials shared by peers.

FR-08: Material Search, Rating & Reporting

- Users shall be able to search and filter materials based on relevant categories.
- Users shall be able to rate/review materials and mark them as helpful.
- Users shall be able to report incorrect, inappropriate, or duplicate materials.

FR-09: Collaborative Question Bank

- Users shall be able to create and contribute questions to channel-specific or general question banks.
- Questions shall support categorization by subject, topic, difficulty, and question type.
- Users shall be able to search, rate, and report questions.

FR-10: Peer Quiz Creation

- Users shall be able to create quizzes using questions from the question bank.
- Quiz creators shall be able to specify parameters such as topic, difficulty, number of questions, and duration.
- Users shall be able to publish quizzes within relevant channels.

FR-11: Quiz Attempt & Evaluation

- Users shall be able to attempt quizzes created by peers or authorized users.
- The system shall automatically evaluate objective questions and calculate scores.
- The system shall store quiz attempts and results for future analysis.

FR-12: LLM-Based Quiz Generation

- The system shall use an LLM to generate questions based on a selected subject, topic, and difficulty.
- The LLM shall generate appropriate options, correct answers, and explanations where applicable.
- Users shall be able to review generated questions before adding them to a quiz/question bank.

FR-13: LLM-Based Adaptive Quiz

- The system shall analyze a user's responses during a quiz.
- The LLM shall dynamically adjust question difficulty based on the user's performance.

- The system shall generate personalized questions while maintaining the selected topic/domain.

FR-14: LLM-Based Answer Evaluation

- The system shall support evaluation of descriptive answers using the LLM.
- The LLM shall evaluate answers based on expected concepts and provide a score/assessment.
- The system shall provide feedback about strengths, weaknesses, and missing concepts.

FR-15: LLM-Based Learning Assistance

- Users shall be able to ask the LLM for explanations of questions or concepts.
- The LLM shall provide examples and simplified explanations where required.
- Users shall be able to request quizzes based on a particular topic or learning area.

FR-16: Channel Chat

- Channel members shall be able to communicate through a channel-based chat.
- The system shall store messages with sender and timestamp information.
- Users shall be able to reply to messages and participate in threaded discussions.

FR-17: Peer-to-Peer Direct Chat

- Users shall be able to initiate private conversations with other users.
- Users shall be able to send and receive direct messages and view conversation history.
- Users shall be able to block or report inappropriate users/messages.

FR-18: Learning Content Sharing through Chat

- Users shall be able to share study materials, questions, and quizzes directly in chat.
- Shared resources shall remain linked to their original content.
- Users shall be able to open shared resources directly from the conversation.

FR-19: Channel-Based LLM Interaction

- Users shall be able to request LLM assistance within a specific channel/domain.
- The system shall generate quizzes using the channel's selected topic or question bank.
- The system may generate quizzes using a combination of **peer questions and LLM-generated questions**.

FR-20: Performance Analysis

- The system shall track scores, attempts, accuracy, and quiz history.

- Users shall be able to view subject-wise and topic-wise performance.
- The system shall display performance trends and identify strong and weak areas.

FR-21: Personalized Recommendations

- The system shall identify topics where a user needs improvement.
- The system shall recommend relevant peer-shared materials, questions, and quizzes.
- Recommendations shall be based on the user's quiz performance and learning activity.

FR-22: Peer Reputation & Contribution

- The system shall track user contributions such as materials, questions, and quizzes.
- Users shall earn reputation based on useful contributions and peer feedback.
- The system may provide contributor rankings or badges.

FR-23: Content Reporting & Moderation

- Users shall be able to report inappropriate, incorrect, duplicate, or misleading content.
- Channel moderators/admins shall be able to review and take action on reports.
- Authorized users shall have additional moderation privileges in Authorized Channels.

FR-24: Notifications

- The system shall notify users about relevant channel activity, replies, and mentions.
- Users shall receive notifications for important quiz or channel updates.
- Users shall be able to manage their notification preferences.

FR-25: Admin Management

- Admins shall be able to manage users, channels, reports, and platform content.
- Admins shall be able to approve/manage Authorized Channels.
- Admins shall be able to suspend users or remove content that violates platform rules.

Non-Functional Requirements

NFR-01: Performance

- The system should respond to normal user operations within an acceptable time.
- Quiz submission and result generation should be fast.
- Search and filtering should provide results efficiently.

NFR-02: Security

- User credentials and personal data shall be securely stored.
- Role-based authorization shall prevent unauthorized actions.
- Users shall only be able to modify resources for which they have permission.

NFR-03: Usability

- The application shall provide a simple and intuitive interface.
- Users should be able to discover channels, materials, questions, and quizzes easily.
- Performance analytics should be presented using clear visualizations.

NFR-04: Reliability

- User-generated materials and quiz results shall be stored reliably.
- The system shall minimize loss of quiz attempts and uploaded content.
- The system should handle temporary failures gracefully.

NFR-05: Scalability

- The system should support increasing numbers of users, channels, materials, questions, and quizzes.
- The architecture should allow additional domains and features to be added later.

NFR-06: LLM Reliability

- The system should validate or allow review of LLM-generated questions before publication.
- The system should provide mechanisms to report incorrect or inappropriate AI-generated content.
- The system should avoid relying solely on LLM output for critical scoring decisions where deterministic evaluation is possible.

NFR-07: Maintainability

- The application should use modular components for channels, chat, quizzes, question banks, analytics, and LLM services.
- The system should allow individual modules to be updated without significantly affecting other modules.

Use Cases

1. User Management

- Register, log in, and manage user profile.
- Track user activity, learning history, streaks, and reputation.

2. Study Material Sharing

- Upload, view, download, search, and organize peer-shared study materials.
- Allow materials to be associated with subjects, domains, topics, or study rooms.

3. Study Room Management

- Create, join, and leave study rooms based on domains, subjects, or topics.
- Share external resources and collaborate with other users within a study room.

4. Quiz Management

- Create and attempt quizzes based on study materials or general topics.
- Configure quiz parameters such as topic, difficulty, and number of questions.

5. Question Bank Management

- Create, contribute, search, and practice questions.
- Generate question banks using shared materials and supplementary online resources.

6. LLM Quiz Generation

- Generate quizzes based on selected topics, materials, or question banks.
- Generate questions, options, answers, and explanations according to specified difficulty.

7. Performance Analysis

- Analyze quiz scores, accuracy, attempts, and topic-wise performance.
- Identify strengths, weaknesses, and topics requiring additional practice.

8. Material Review & Recommendation

- Use the LLM to review study materials and provide suggestions.
- Recommend relevant materials based on user performance, interests, and learning needs.

9. Personalized Material List

- Allow users to save and organize materials they want to study later.

- Allow users to add LLM-recommended or peer-recommended resources to their personal list.

10. Peer Review & Rating

- Allow users to rate and review study materials and questions.
- Use peer feedback to identify useful, reliable, and high-quality content.

11. Peer Validation & Recommendation

- Allow users to validate or report the quality and accuracy of peer-generated content.
- Recommend highly rated and relevant peer content to other users.

12. Streak & Reputation Management

- Track user learning activity and maintain learning streaks.
- Award reputation points and badges based on contributions, participation, and achievements.

13. Content Moderation

- Allow users to report inappropriate, inaccurate, duplicate, or misleading content.
- Allow admins/moderators to review reports and remove or restrict problematic content.

Actors

1. Student / User

- The primary actor who consumes and contributes to the platform.
- Can share materials, create/attempt quizzes, join study rooms, participate in discussions, and review peer content.
- Can receive LLM-based recommendations and track personal progress, streaks, and reputation.

2. Admin / Moderator

- Manages users, study rooms, reported content, and platform-wide activities.
- Reviews inappropriate, inaccurate, or reported materials and questions.
- Manages authorized content and handles moderation-related activities.

3. LLM / AI Service

- Generates quizzes and analyzes quiz performance.
- Reviews study materials and provides summaries, suggestions, and personalized recommendations.
- Processes uploaded videos to generate study materials, summaries, and potentially questions.

4. External Resource / Online Source

- Provides additional educational resources and questions that can be incorporated into the platform.
- Acts as a source for supplementary study material and question-bank content.